

ShanghaiTech University
School of Information Science and Technology

EE115 A Lab Experiments

Experiment 7: Single Stage CS Amplifier Lab Worksheet

Contents

1. Pre-Lab.....	1
1.1. Attenuation Network	1
1.2. Single Stage CS MOSFET Amplifier	1

1. Pre-Lab

Single Stage CS MOSFET Amplifier

Explain the function of R_s :

Small Signal Circuit:

Expression of middle band gain A_v : _____

Expression of output impedance of the amplifier R_{out} : _____

Expression of high cutoff frequency f_H : _____

Table 1: Component Design

Component Design	Hand Calculation	Multisim Simulation
R_{g1}		
R_{g2}		
R_d		
R_s		

Table 2: Performance Verification

Performance	Hand Calculation	Multisim Simulation
Middle Band Gain(A_{mid})		
High Cutoff Frequency(f_H)		
Output Swing(SW)		
Total Power Consumption(P_{total})		

Plot Amplifier Gain (in dB20) v.s. Frequency: